Year 2 Statistics and Mechanics

Chapter 1

Regression, correlation and hypothesis testing

Practice Booklet

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1.1 Exponential models

Practice Book, printed page 1.

1. Data are coded using $Y = \log y$ and $X = x$ to give a linear relationship.
The equation of the regression line for the coded data is $Y = 0.7 + 1.2X$.

a. State whether the relationship between y and x is of the form $y = ax^n$ or $y = kb^x$.

b. Write down the relationship between y and x and find the values of the constants.

Hint

Use the change of variable to find an expression for $\log y$ in terms of x. Then use the fact that $10^{\log y} = y$ and the laws of indices to find the form of the relationship. ← Pure Year 1, Section 14.8

2. Data are coded using $Y = \log y$ and $X = \log x$ to give a linear relationship.
The equation of the regression line for the coded data is $Y = 1.5 + 0.8X$.

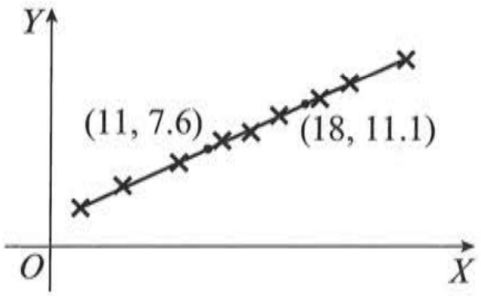
a. State whether the relationship between y and x is of the form $y = ax^n$ or $y = kb^x$.

b. Write down the relationship between y and x and find the values of the constants.

Hint

Use the change of variable to find an expression for $\log y$ in terms of $\log x$.

3. P The scatter diagram shows the relationship between two sets of coded data, X and Y, where $X = x$ and $Y = \log y$. The regression line of Y on X is shown, and passes through the points $(11, 7.6)$ and $(18, 11.1)$.



The relationship between the original data sets is modelled by an equation of the form $y = kb^x$. Find the values of k and b.

4. P The table shows the populations of the ten most populated districts of England. R is the population ranking and P is the population in 1000s.

R	1	2	3	4	5	6	7	8	9	10
P	1137	785	578	561	546	535	524	496	492	459

a. Plot a scatter diagram showing $\log P$ against $\log R$.

b. Comment on the correlation between $\log P$ and $\log R$.

c. State whether your answer to part **b** supports that fact that the original data can be modelled by a relationship of the form $P = aR^n$.

The data are coded using the change of variable $x = \log R$, $y = \log P$.
The regression line of y on x is found to be $y = 3.00 - 0.355x$.

d. Find an equation for P in terms of R, giving your answer in the form $P = aR^n$, where a and n are constants to be found.

5. P Specific humidity can be expressed as the grams of water vapour per kilogram of atmospheric gas. The table shows the temperature, in °C, and the specific humidities recorded at different atmospheric temperatures.

Temperature, t (°C)	−40	−30	−20	−10	0	10	20	30	40
Specific humidity, h (g/kg)	0.1	0.3	0.75	2	3.5	7	14	26.5	47

a. Plot a scatter diagram showing $\log h$ against t.

b. Comment on the correlation between $\log h$ and t.

c. Use your answer to part **a** to explain why an equation of the form $h = kb^t$, where k and b are constants, is likely to be a good model for the relationship between t and h.

d. The regression line of $\log h$ on t is given as $\log h = 0.494 + 0.0327t$. Determine the values of the constants k and b in the equation given in part **c**.

6. Data are collected on the amounts of water, W million gallons per day, used to generate electricity, E million kilowatt hours, from a sample of hydroelectric generators. The data are coded using $x = \log W$ and $y = \log E$. It is found that a linear relationship exists between x and y and that the equation of the regression line of y on x is $y = 1.02x - 1.22$.
Find an equation for E in terms of W, giving your answer in the form $E = aW^n$, where a and n are constants to be found.

7. Data are collected on the lengths, l m, and masses, m g, of a sample of rainbow trout from a river. The data are coded using $x = l$ and $y = \log m$. It is found that a linear relationship exists between x and y and that the equation of the regression line of y on x is $y = 3.27x + 1.50$.
Find an equation to describe the relationship between m and l, giving your answer in the form $m = kb^l$, where k and b are constants to be found.

8. E/P The table shows some data collected on the amounts of nitrogen, kg per hectare, added to an agricultural field and the harvest of grain, kg per hectare.

Nitrogen input, N (kg/ha)	101	142	169	255	325	408	481	564	601	635
Harvest of grain, H (kg/ha)	8600	10900	12500	16200	19100	22800	24400	27600	28700	29200

The data are coded using $x = \log N$ and $y = \log H$.
The regression line of y on x is found to be $y = 2.6 + 0.67x$.

a. Given that the data can be modelled by an equation of the form $y = ax^n$, where a and n are constants, find the values of a and n. (3 marks)

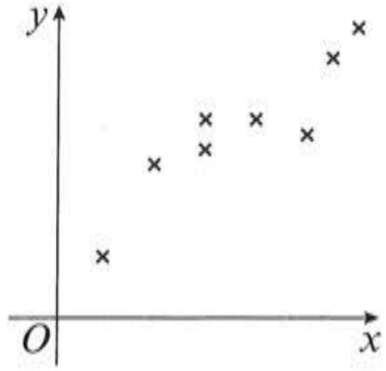
b. Explain why this model is not reliable for estimating the harvest of a field with a nitrogen input of 900 kg/ha. (1 mark)

1.2 Measuring correlation

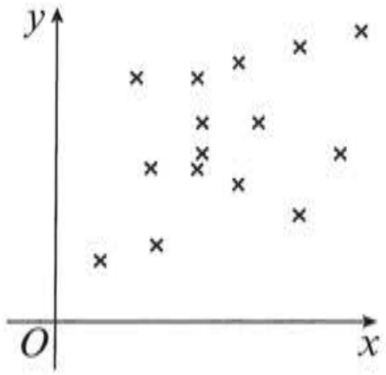
Practice Book, printed page 3.

1. Suggest a value of r for each of these scatter diagrams.

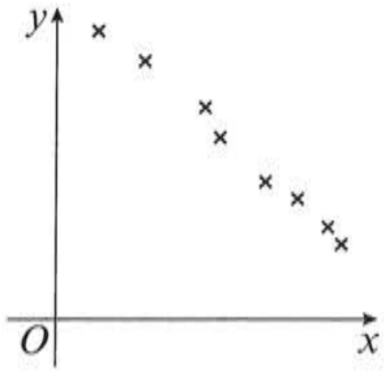
a



b



c



Hint

The product moment correlation coefficient, r, measures linear correlation. The closer r is to -1 or 1, the stronger the negative or positive linear correlation, respectively.

2. The table shows 10 observations from a bivariate data set.

x	36	14	72	6	54	94	23	66	87	41
y	78	67	281	29	210	312	84	237	305	160

a. State what is measured by the product moment correlation coefficient.

b. Find the value of the product moment correlation coefficient between x and y.

Hint

Use your calculator to find the product moment correlation coefficient. Give your answer to 4 decimal places.

3. The table shows the velocity of a river current at different depths within the channel.

Depth, d (m)	0.79	1.4	2.62	0.61	1.8	3.41	1.01	2.23	0.21	3.23
Velocity, v (m s^{-1})	0.43	0.42	0.18	0.34	0.35	0.07	0.42	0.28	0.47	0.12

a. Use your calculator to find the value of the product moment correlation coefficient between d and v.

b. Use your answer to part **a** to describe the correlation between the depth and velocity of the river based on this sample.

4. P The table shows the populations of the ten most populated districts of England. R is the population ranking and P is the population in 1000s.

R	1	2	3	4	5	6	7	8	9	10
P	1137	785	578	561	546	535	524	496	492	459

The data are coded using $x = \log R$, $y = \log P$.
Calculate the product moment correlation coefficient for the coded data.

5. E/P The table shows the masses of household waste, W, for ten towns with population P.

Population, P (1000s)	87	9	73	62	7	44	53	98	29	35
Waste, W (tonnes 1000s)	95	19	58	41	14	37	47	82	24	26
$\log W$										

The data are coded using $x = P$ and $y = \log W$.

a. Copy and complete the table showing the values of $\log W$. (2 marks)

b. Calculate the product moment correlation coefficient for the coded data. (1 mark)

c. With reference to your answer to part **b**, state whether an exponential model is a good fit for these data. (2 marks)

The equation of the regression line of y on x is found to be $y = 1.15 + 0.0085x$.

d. Find an expression for W in terms of P, giving your answer in the form $W = ab^P$, where a and b are constants to be found. (3 marks)

6. E/P The table shows the engine sizes, E litres, and fuel economies, M miles per gallon, for a sample of ten cars.

Engine size, E (litres)	3.2	3.2	2.8	2.5	2.8	1.8	5	6	2.3	3.5
Fuel economy, M (mpg)	24	26	26	27	26	29	21	19	30	23
$\log E$										
$\log M$										

The data are coded using $x = \log E$ and $y = \log M$.

a. Copy and complete the table showing the values of $\log E$ and $\log M$. (3 marks)

b. Calculate the product moment correlation coefficient for the coded data. (1 mark)

c. With reference to your answer to part **b**, state whether a model of the form $y = kx^n$, where k and n are constants, is a good fit for these data. (2 marks)

The equation of the regression line of y on x is found to be $y = 1.59 - 0.383x$.

d. Determine the values of k and n. (3 marks)

7. E From the large data set, the daily mean air temperatures, t °C, and daily mean visibilities, v Dm, were recorded for Camborne on seven consecutive days in August 1987.

Temp, t (°C)	16.8	16.7	15.8	14.4	13.9	14.0	13.8
Visibility, v (Dm)	2100	700	2500	6300	6200	4700	3900

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a. Calculate the product moment correlation coefficient for these data. (1 mark)

b. With reference to your answer to part **a**, comment on the suitability of a linear regression model for these data. (2 marks)

8. E From the large data set, the daily maximum gusts, g kn and daily mean visibilities, v Dm, were recorded for Leuchars on seven consecutive days in July 1987.

Maximum gust, g (kn)	19	18	17	24	16	n/a	27
Visibility, v (Dm)	3100	3100	3100	4800	5100	1300	3200

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a. State the meaning of n/a in the table. (1 mark)

b. Calculate the product moment correlation coefficient for these data. State how you have dealt with the data entry marked n/a. (2 marks)

Jamie states that because the value of r is close to 0, there is no relationship between daily maximum gust and daily mean visibility.

c. Comment on Jamie's statement. (1 mark)

1.3 Hypothesis testing for zero correlation

Practice Book, printed page 5.

1. A population of students each took two different tests. A sample of 30 students was taken from the population and their scores on the two tests were recorded. A product moment correlation coefficient of 0.4125 was calculated.
Test whether or not this shows evidence of positive correlation between the test scores:

a. at the 5% level

b. at the 1% level.

Hint

You need to use a one-tailed test with $H_0 : \rho = 0$, $H_1 : \rho > 0$.
Use the table of critical values on page 92. ← Year 1, Section 7.1

2. A biologist takes a sample of 12 otters and records their masses and lengths. The product moment correlation coefficient of the data is found to be 0.62.
Test whether or not this shows evidence of correlation between the mass and length:

a. at the 10% level

b. at the 2% level.

Hint

You need to use a two-tailed test with $H_0 : \rho = 0$, $H_1 : \rho \neq 0$. Remember to halve the probability in each tail.

3. The table shows the engine sizes, E litres, and fuel economies, M miles per gallon, for a sample of six cars.

Engine size, E (litres)	3.8	5.7	2.4	3.4	2.2	2.4
Fuel economy, M (mpg)	30	28	30	28	34	33

a. Calculate the product moment correlation coefficient for these data, correct to 3 significant figures.

b. For these data, test $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$, using a 5% significance level.

4. P Each year for eight years, an oil company recorded the number of barrels of oil it produced that year and the total number of active wells it was using by the end of the year.

Year	2010	2011	2012	2013	2014	2015	2016	2017
Wells	46	157	237	337	419	501	601	657
Barrels of oil (millions)	5	21	17	20	25	28	24	30

The company believes that the number of active wells and the number of barrels of oil produced in a year are positively correlated and can be modelled by a linear relationship. Using a 1% significance level, test to see if there is evidence that the company's beliefs are correct.

5. E A city council collects data on the population densities in different areas of the city, in people per hectare, and the distances of those areas from the city centre, in km. It calculates the product moment correlation coefficient between the two sets of data and finds it to be -0.51.
Given that the council collected data from 24 sample areas, test, at the 1% level of significance, the claim that there is a negative correlation between population density and the distance from the city centre. State your hypotheses clearly. (3 marks)

6. P A zoologist believes that there is positive correlation between the lengths and masses of rainbow trout in a river. She decides to carry out a hypothesis test to see if there is evidence for her claim. She takes a random sample of 12 rainbow trout and finds that the correlation coefficient is 0.7.
Given that this result provided the zoologist with sufficient evidence to reject her null hypothesis, suggest the least possible significance level for the zoologist's test.

7. P Data on the daily mean temperature and the rainfall is taken from the large data set for Perth in August 2015. A meteorologist finds that the product moment correlation coefficient for these data is -0.833. Given that the researcher tests for negative correlation at the 0.5% level of significance, and concludes that the value is significant, find the smallest possible sample size.

8. E The Mercalli scale and the Richter scale are both used to measure earthquakes. The Mercalli scale is based on structural damage to buildings and the Richter scale is based on the amplitude of the ground vibrations. The table shows data from eight earthquakes.

Mercalli intensity, M	9	7	8	6	5	8	6	8
Richter magnitude, R	6.3	5.3	6.0	5.5	4.0	5.4	4.7	7.1

a. Calculate, to 3 decimal places, the product moment correlation coefficient between M and R. (1 mark)

It is suggested that there is linear correlation between the structural damage and ground vibrations caused by an earthquake.

b. Test this suggestion at the 1% significance level, stating your null and alternative hypotheses clearly. (3 marks)

9. E An employee at a weather centre believes that there is positive correlation between the daily mean windspeed, w knots, and the daily maximum gust, g knots.
The employee takes a sample of data from Heathrow in May 2015.

Maximum windspeed, w (kn)	9	10	14	12	14	9	13	5
Maximum gust, g (kn)	21	22	41	24	30	25	29	19

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a. Calculate the product moment correlation coefficient for these data. (1 mark)

b. Test, at the 2.5% level of significance, the employee's claim. State your hypotheses clearly. (3 marks)

Problem solving

Bronze

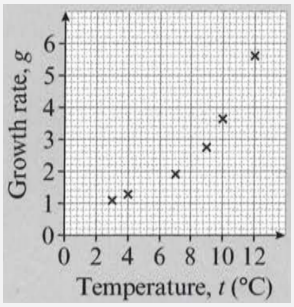
The table shows some data collected on body weights (in kilograms) and corresponding brain weights (in grams) for seven terrestrial mammals.

Body weight, x (kg)	53.6	0.01	250	519	3.38	0.55	1.45
Brain weight, y (g)	450	0.48	510	680	42.5	3.2	14.9

Test, at the 0.5% level of significance, whether these results show evidence of a linear relationship between x and y. State your hypotheses clearly. (4 marks)

Silver

A biologist collected data on the temperature, $t\,^{\circ}\mathrm{C}$, of a colony of bacteria and its growth rate g. The data are shown on the scatter diagram.



a. With reference to the scatter diagram, explain why a linear regression model may not be suitable for the relationship between t and g. (1 mark)

The table shows the data collected.

Temperature, t (°C)	3	4	7	9	10	12
Growth rate, g	1.09	1.28	1.91	2.75	3.64	5.61

The data are coded using $x = t$ and $y = \log g$.

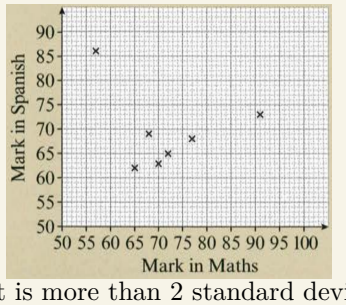
b. Stating your hypothesis clearly, test, at the 5% significance level, whether the product moment correlation coefficient for the coded data is greater than 0. (5 marks)

Gold

Seven students take a maths test and a Spanish test. The maths test is marked out of 100 and the Spanish test is marked out of 75.
The table shows the marks attained by the seven students in both tests.

Student	A	B	C	D	E	F	G
Maths	70	72	68	65	57	77	91
Spanish	63	65	69	62	86	68	73

A graph of the data is shown.



An outlier is defined as a data value that is more than 2 standard deviations from the mean.

a. Show that the data for student E is an outlier. (3 marks)

A teacher claims that the results of the two tests are linearly correlated and wishes to use a hypothesis test to support this claim.

b. By carrying out suitable tests at the 10% significance level, determine whether the removal of the data for student E will affect the teacher's conclusion. You should state the hypotheses for any tests, and label your working clearly. (7 marks)

c. Give a reason to support the removal of the data for student E, and hence write a short conclusion based on your answer to part **b**. (2 marks)

Selected exam question bank questions

Question 2

A chemist carried out an experiment seven times, and recorded the masses of the reactant and product each time.

Mass of reactant, m (grams)	3	5	7	10	15	20	30
Mass of product, p (grams)	2.7	4.1	5.2	6.8	9.1	11.3	15.4

The data are believed to be modelled by a relationship of the form $p = am^k$, where a and k are constants. The data are coded using $x = \log m$ and $y = \log p$.

The equation of the regression line of y on x is found to be $y = 0.081 + 0.749x$.

a. Find the value of a and the value of k. (2 marks)

b. Give a reason why this model would not be suitable for estimating the mass of the product in a similar experiment with 50 grams of reactant. (1 mark)

Question 4

Data on the daily total sunshine, s hours, and daily mean visibility, v Dm, is taken from the large data set for Leeming in July 2015.

Sunshine, s (hours)	5.9	14.5	4.6	6.8	6.5	10.7
Visibility, v (Dm)	3200	3400	1900	2200	2500	3600

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a. Calculate the product moment correlation coefficient for these data. (1 mark)

b. Test, at the 5% level of significance, the claim that there is positive correlation between the daily total sunshine and visibility. State your hypotheses clearly. (3 marks)

Question 7

From the large data set, the daily mean air temperatures, $t\,^\circ$C, and the daily mean windspeeds, w knots, were recorded for a location in the northern hemisphere during the last week of October 2015.

Temperature, t (°C)	23.0	23.0	22.6	24.1	22.2	20.4	19.0
Windspeed, w (kn)	6.5	7.6	6.3	6.0	6.0	3.5	3.1

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a. Using your knowledge of the large data set, suggest a location where the data was taken. (1 mark)

The corresponding data for Perth are shown in this table.

Temperature, t (°C)	19.7	21.3	21.0	20.6	21.0	20.8	18.8
Windspeed, w (kn)	10.1	12.3	9.9	7.4	6.7	7.8	8.5

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b. Calculate the product moment correlation coefficient for these data. (1 mark)

c. With reference to your answer to part **b**, comment on the suitability of a linear regression model for these data. (2 marks)

Question 10

Data are collected on the amounts of water, W million gallons per day, used to generate electricity, E millions of kilowatt hours, from a sample of eight different hydroelectric generators.

Water, W	24000	109000	19000	14000	67000	13000	3000	46000
Electricity, E	1090	3040	840	2190	10670	830	1600	1980

a. Calculate the product moment correlation coefficient for these data, correct to 3 significant figures. (1 mark)

b. Test, at the 10% significance level, whether these results show evidence of a positive linear relationship between W and E. State your hypotheses clearly. (3 marks)

Data are then collected on the amounts of water, W million gallons per day, used to generate electricity, E millions of kilowatt hours, from a sample of eight different days for one hydroelectric generator.

Water, W	24000	32000	14000	22000	37000	43000	27000	19000
Electricity, E	1090	1640	670	870	2380	2460	1950	940

c. Calculate the product moment correlation coefficient for these data, to 3 s.f. (1 mark)

d. Test, at the 10% significance level, whether these results show evidence of a positive linear relationship between W and E. State your hypotheses clearly. (1 mark)

e. Give a possible reason for the difference in the results of the hypothesis tests in parts **b** and **d**. (1 mark)

Question 14

A geologist collected these data on the diameters of solid particles in a river and the speeds of the current required to keep the particles in suspension.

Diameter of particle, d (mm)	0.2	1.3	5	11	20	45	80	160
Speed of current, v (m s^{-1})	0.1	0.25	0.5	0.75	1	1.5	2.5	3.5

The data are coded using $x = d$ and $y = \log v$.

The equation of the regression line of y on x is found to be $y = 0.0074x - 0.412$.

a. Find an expression for v in terms of d, giving your answer in the form $v = kb^d$, where k and b are constants to be found. (3 marks)

b. Calculate the product moment correlation coefficient for the coded data. (3 marks)

The geologist believes that there is positive linear correlation between the coded data.

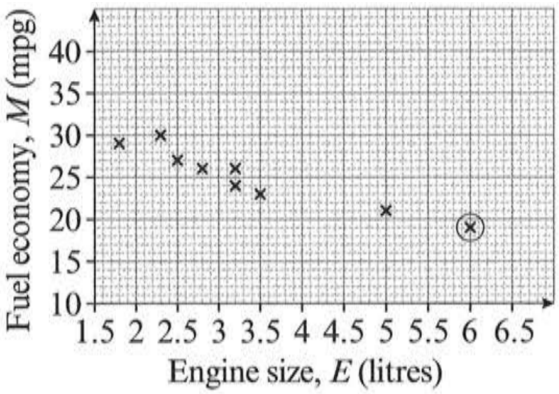
c. Given that the data provided the geologist with sufficient evidence to reject the null hypothesis, suggest the least possible significance level for the geologist's test. (1 mark)

Question 17

A car manufacturer records the engine sizes, E litres, and fuel economies, M miles per gallon, for a sample of ten cars.

Engine size, E (litres)	3.2	3.2	2.8	2.5	2.8	1.8	5.0	6.0	2.3	3.5
Fuel economy, M (mpg)	24	26	26	27	26	29	21	19	30	23

The data are represented on a scatter diagram.



Values greater than $Q_3 + (Q_3 - Q_1)$ or less than $Q_1 - (Q_3 - Q_1)$ are considered to be outliers.

a. Show that the point circled on the diagram is an outlier. (4 marks)

b. Suggest a reason why you might elect not to remove this outlier from your calculations. (1 mark)

c. Calculate the product moment correlation coefficient for all the data. (1 mark)

d. Suggest a possible reason for this correlation. (1 mark)

e. With reference to your answer to part **c**, comment on the suitability of a linear regression model for these data. (1 mark)

The manufacturer calculated the linear regression equation as $M = 2.53E + 33.5$.

f. Explain why the manufacturer's equation is incorrect. (1 mark)

Answers

1.1 Exponential models

1.

a. $y = kb^x$

b. $k = 5.01$, $b = 15.8$ (3 s.f.)

2.

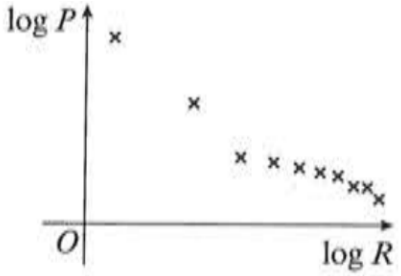
a. $y = ax^n$

b. $a = 31.6$ (3 s.f.), $n = 0.8$

3. $k = 126$, $b = 3.16$ (3 s.f.)

4.

a.



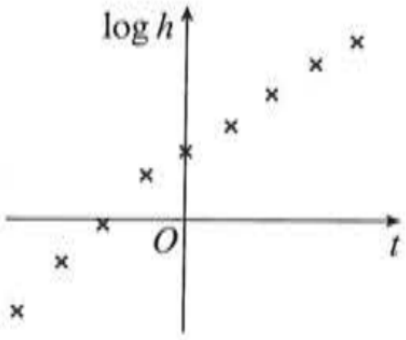
b. Strong negative correlation

c. Yes; the variables show a linear relationship when $\log P$ is plotted against $\log R$.

d. $P = 1000R^{-0.355}$

5.

a.



b. Strong positive correlation

c. The variables show a linear relationship when $\log h$ is plotted against t.

d. $k = 3.12$, $b = 1.08$ (3 s.f.)

6. $E = 0.0603W^{1.02}$, $a = 0.0603$, $n = 1.02$ (3 s.f.)

7. $k = 31.6$, $b = 1860$ (3 s.f.)

8.

a. $a = 398$ (3 s.f.), $n = 0.67$

b. 900 kg/ha is outside the range of the data (extrapolation).

1.2 Measuring correlation

1.

a. Approximately 0.7

b. Approximately 0.3

c. Approximately -0.9

2.

a. The type and strength of linear correlation between x and y

b. 0.9783

3.

a. -0.9355

b. There is a strong negative correlation. The parts of the channel with larger depth have a lower current velocity.

4. -0.954 (3 s.f.)

5.

a.

P	87	9	73	62	7	44	53	98	29	35
W	95	19	58	41	14	37	47	82	24	26
$\log W$	1.98	1.28	1.76	1.61	1.15	1.57	1.67	1.91	1.38	1.41

b. 0.975 (3 s.f.)

c. There is strong positive linear correlation with data in the form $\log W$ against P, which suggests an exponential model is a good fit.

d. $a = 14.1$, $b = 1.02$ (3 s.f.)

6.

a.

E	3.2	3.2	2.8	2.5	2.8	1.8	5	6	2.3	3.5
M	24	26	26	27	26	29	21	19	30	23
$\log E$	0.505	0.505	0.447	0.398	0.447	0.255	0.699	0.778	0.362	0.544
$\log M$	1.38	1.41	1.41	1.43	1.41	1.46	1.32	1.28	1.48	1.36

b. -0.962 (3 s.f.)

c. There is strong negative linear correlation with data in the form $\log M$ against $\log E$, which suggests a model of the form $y = kx^n$ is a good fit.

d. $k = 38.9$ (3 s.f.), $n = -0.383$

7.

a. -0.852

b. -0.852 is close to -1, so the data values show a strong to moderate negative correlation. A linear regression model is suitable for these data.

8.

a. Data on daily maximum gust is not available for this day.

b. -0.0549. The n/a data entry is omitted when calculating r.

c. Jamie's statement is not valid because there may be a non-linear relationship between the two variables or the sample size is too small to draw this conclusion.

1.3 Hypothesis testing for zero correlation

1.

a. $H_0 : \rho = 0$, $H_1 : \rho > 0$, critical value $= 0.3061 < 0.4125$.
Reject H_0: there is reason to believe at the 5% level of significance that there is a correlation between the scores.

b. $H_0 : \rho = 0$, $H_1 : \rho > 0$, critical value $= 0.4226 > 0.4125$.
Accept H_0: there is no evidence of correlation between the two scores at the 1% level of significance.

2.

a. $H_0 : \rho = 0$, $H_1 : \rho \neq 0$, critical value $= 0.4973 < 0.62$.
Reject H_0: there is reason to believe at the 10% level of significance that there is a correlation between the mass and the length of the otters.

b. $H_0 : \rho = 0$, $H_1 : \rho \neq 0$, critical value $= 0.6581 > 0.62$. Accept H_0: there is no evidence of correlation between the mass and the length of the otters at the 2% level of significance.

3.

a. -0.728

b. $H_0 : \rho = 0$, $H_1 : \rho \neq 0$, critical value $= 0.8114$. Accept H_0: there is no evidence of correlation between the engine size and fuel economy at the 5% level of significance.

4. $H_0 : \rho = 0$, $H_1 : \rho > 0$, critical value $= 0.7887$, $r = 0.861$.
Reject H_0: there is evidence of linear correlation at the 1% level of significance. There is evidence to suggest the company is correct.

5. $H_0 : \rho = 0$, $H_1 : \rho < 0$, critical value $= 0.4716$.
Reject H_0: there is evidence of negative correlation at the 1% level of significance. There is evidence to suggest the council is correct.

6. 1%

7. 9

8.

a. 0.812

b. $r = 0.812$, $H_0 : \rho = 0$, $H_1 : \rho > 0$
Critical value $= 0.7887 < 0.812$.
Reject H_0: there is evidence of positive correlation at the 1% level of significance.

9.

a. 0.786

b. $H_0 : \rho = 0$, $H_1 : \rho > 0$, critical value $= 0.7067$.
Reject H_0: there is evidence of linear correlation at the 2.5% level of significance.

Problem solving

Bronze

$r = 0.870$ (3 s.f.). $H_0 : \rho = 0$, $H_1 : \rho > 0$

Critical value $= 0.8745 > 0.870$.

Accept H_0: there is no evidence of positive correlation between x and y at the 0.5% level of significance.

Silver

a. The data points do not fit a straight line.

b. $r = 0.959$ (3 s.f.). $H_0 : \rho = 0$, $H_1 : \rho > 0$
Critical value $= 0.7293 < 0.959$.
Reject H_0: there is evidence of positive correlation at the 5% level of significance.

Gold

a. E.g. using Spanish results, mean $= 69.4$ and s.d. $= 7.61$, so $86 >$ mean $+ 2 \times$ s.d.

b. All data:
$r = -0.2313$. $H_0 : \rho = 0$; $H_1 : \rho \neq 0$, $n = 7$, 5% in each tail gives critical value $= -0.6694 < -0.2313$.
Accept H_0: there is no evidence of correlation between the marks, so claim would be incorrect.
Removing point E:
$r = 0.8158$. $H_0 : \rho = 0$; $H_1 : \rho \neq 0$, $n = 6$, 5% in each tail gives critical value $= 0.7293 < 0.8158$.
Reject H_0: there is evidence of correlation between the marks, so claim would be correct.
The removal of the outlier would affect the conclusion.

c. Point E is likely to be an incorrect data entry (as the Spanish score is greater than the maximum number of marks available on the test, 75). It would be reasonable to remove this data point and conclude that there is evidence of correlation between the marks in the two tests.

Selected exam question bank answers

Question 2

a. $a = 1.21$ (3 s.f.), $k = 0.749$

b. 50 grams is outside the range of the data (extrapolation).

Question 4

a. 0.731 (3 s.f.)

b. $H_0 : \rho = 0$, $H_1 : \rho > 0$, critical value $= 0.7293 < 0.731$.
Reject H_0: there is evidence of positive correlation at the 5% level of significance, so claim is correct.

Question 7

a. Jacksonville or Beijing

b. 0.133 (3 s.f.)

c. The data shows a very weak positive correlation so a linear model may not be best. There may be other variables affecting the relationship or a different model might be a better fit.

Question 10

a. 0.496

b. $H_0 : \rho = 0$, $H_1 : \rho > 0$, critical value $= 0.5067$.
Accept H_0: there is no evidence of linear correlation at the 10% level of significance.

c. 0.9370

d. $H_0 : \rho = 0$, $H_1 : \rho > 0$, critical value $= 0.5067$.
Reject H_0: there is evidence of linear correlation at the 10% level of significance.

e. Weaker positive correlation in part **b** due to different efficiencies of the different generators.

Question 14

a. $k = 0.387$, $b = 1.02$ (3 s.f.)

b. $r = 0.799$ (3 s.f.)

c. 1% (critical value $= 0.7887$)

Question 17

a. Using data for engine size,
$Q_3 + (Q_3 - Q_1) = 3.875 + 1.425 = 5.3 < 6$

b. It follows the trend of the data, or it is a reasonable engine capacity for a large car.

c. -0.943

d. Cars with larger engines are larger and therefore less fuel efficient.

e. Strong negative correlation, therefore a linear model is likely to be suitable.

f. Negative correlation, so expect b to be negative.